

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

32,641
held-out plate appearances

3.05%
held-out HR rate

0.5999
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-27, scored on the 32,641 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5836	1.55×	0.02952	0.13556	0.00462
KNN comps Prior is the pooled rate of the nearest comparables.	0.5999	1.52×	0.02949	0.13463	0.00494
SVR Support-vector regression on the same feature matrix.	0.5969	1.48×	0.02953	0.13522	0.00618
Random forest Forest regression; a learned similarity metric.	0.5958	1.47×	0.02952	0.13504	0.00551
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5999	1.52×	0.02949	0.13463	0.00494
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5999	1.52×	0.02949	0.13463	0.00494
Linear regression Ordinary least squares on the same features.	0.5965	1.47×	0.02951	0.13498	0.00436
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5976	1.54×	0.02953	0.13512	0.00680
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.46×	0.02951	0.13496	0.00557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5971	1.45×	0.02953	0.13503	0.00625

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,547 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5526	1.44×	0.02288	0.11153	0.00903
KNN comps Prior is the pooled rate of the nearest comparables.	0.5744	1.20×	0.02296	0.11191	0.01517
SVR Support-vector regression on the same feature matrix.	0.5731	1.32×	0.02313	0.11507	0.01888
Random forest Forest regression; a learned similarity metric.	0.5821	1.08×	0.02314	0.11331	0.01866
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5744	1.20×	0.02296	0.11191	0.01517
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5744	1.20×	0.02296	0.11191	0.01517
Linear regression Ordinary least squares on the same features.	0.5768	0.60×	0.02307	0.11384	0.01888
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5769	0.48×	0.02314	0.11364	0.02055
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5743	1.32×	0.02303	0.11288	0.01760
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5886	1.08×	0.02315	0.11327	0.01986

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Matt Olson 16.1%, Fernando Tatis Jr. 15.8%, Ronald Acuña Jr. 15.5%, Jackson Chourio 14.5%, Drake Baldwin 13.9%, Jackson Merrill 13.8%
Flat league	5/6	Fernando Tatis Jr. 15.7%, Ronald Acuña Jr. 14.8%, Matt Olson 14.5%, Jackson Chourio 14.5%, Drake Baldwin 13.6%, Manny Machado 12.7%
KNN comps	6/6	Fernando Tatis Jr. 16.1%, Matt Olson 15.7%, Ronald Acuña Jr. 15.5%, Jackson Chourio 14.9%, Drake Baldwin 14.1%, Jackson Merrill 13.7%
SVR	6/6	Matt Olson 16.2%, Fernando Tatis Jr. 15.9%, Ronald Acuña Jr. 15.5%, Jackson Chourio 15.0%, Drake Baldwin 14.3%, Jackson Merrill 13.7%
Random forest	6/6	Matt Olson 16.1%, Fernando Tatis Jr. 15.7%, Ronald Acuña Jr. 15.3%, Jackson Chourio 15.1%, Drake Baldwin 14.7%, Jackson Merrill 13.7%
Core (KNN+SVR)	6/6	Fernando Tatis Jr. 16.1%, Matt Olson 15.7%, Ronald Acuña Jr. 15.5%, Jackson Chourio 14.9%, Drake Baldwin 14.1%, Jackson Merrill 13.7%
Forest blend	6/6	Fernando Tatis Jr. 16.1%, Matt Olson 15.7%, Ronald Acuña Jr. 15.5%, Jackson Chourio 14.9%, Drake Baldwin 14.1%, Jackson Merrill 13.7%
Linear regression	6/6	Fernando Tatis Jr. 16.3%, Matt Olson 15.8%, Ronald Acuña Jr. 15.3%, Jackson Chourio 14.9%, Drake Baldwin 14.3%, Jackson Merrill 13.7%
Logistic regression	6/6	Fernando Tatis Jr. 16.4%, Matt Olson 16.4%, Ronald Acuña Jr. 15.3%, Jackson Chourio 14.9%, Drake Baldwin 14.2%, Jackson Merrill 14.0%
Neural net (torch)	6/6	Fernando Tatis Jr. 16.4%, Matt Olson 16.1%, Ronald Acuña Jr. 15.4%, Jackson Chourio 15.0%, Drake Baldwin 14.1%, Jackson Merrill 13.9%
XGBoost	6/6	Matt Olson 16.1%, Fernando Tatis Jr. 15.7%, Ronald Acuña Jr. 15.3%, Jackson Chourio 15.2%, Drake Baldwin 14.7%, Jackson Merrill 13.9%

Hitter by hitter

The top 40 bats by the served probability.

MLB Season Performance Analysis - Q3 2024											
HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO	
1Matt Olson	slate 16.1%	14.5%	15.7%	16.2%	16.1%	15.7%	15.7%	15.8%	16.4%		
2Fernando Tatis Jr.	slate 15.8%	15.7%	16.1%	15.9%	15.7%	16.1%	16.1%	16.3%	16.4%		
3Ronald Acuña Jr.	slate 15.5%	14.8%	15.5%	15.5%	15.3%	15.5%	15.5%	15.3%	15.3%		
4Jackson Chourio	slate 14.5%	14.5%	14.9%	15.0%	15.1%	14.9%	14.9%	14.9%	14.9%		
5Drake Baldwin	slate 13.9%	13.6%	14.1%	14.3%	14.7%	14.1%	14.1%	14.3%	14.2%		
6Jackson Merrill	slate 13.8%	12.4%	13.7%	13.7%	13.7%	13.7%	13.7%	13.7%	14.0%		
7Escar Hernandez	slate 13.3%	12.6%	13.4%	13.5%	13.7%	13.4%	13.4%	13.6%	13.6%		
8Manny Machado	slate 13.1%	12.7%	13.2%	13.6%	13.3%	13.2%	13.2%	13.4%	13.5%		
9William Contreras	slate 12.5%	12.3%	12.5%	12.3%	12.3%	12.5%	12.5%	12.4%	12.4%		
10Shohei Ohtani	slate 12.3%	11.0%	12.4%	12.4%	12.5%	12.4%	12.4%	12.3%	12.6%		
11Jake Bauers	slate 12.0%	11.2%	12.3%	12.2%	12.2%	12.3%	12.3%	12.1%	12.0%		
12Will Smith	slate 11.7%	11.5%	11.7%	11.8%	12.3%	11.7%	11.7%	12.3%	12.1%		
13Garrett Mitchell	slate 11.6%	11.1%	11.6%	10.2%	10.8%	11.6%	11.6%	11.0%	10.9%		
14Mookie Betts	slate 11.3%	11.2%	11.1%	11.8%	11.1%	11.1%	11.1%	11.4%	11.5%		
15Austin Riley	slate 11.2%	10.3%	10.9%	10.9%	10.7%	10.9%	10.9%	10.9%	11.0%		
16Ty France	slate 10.8%	10.6%	11.0%	10.9%	11.1%	11.0%	11.0%	11.0%	11.0%		
17Brice Turang	slate 10.7%	11.0%	10.5%	10.8%	10.4%	10.5%	10.5%	10.1%	10.1%		
18Sean Murphy	slate 10.4%	10.0%	10.6%	10.0%	10.7%	10.6%	10.6%	10.9%	10.2%		
19Michael Harris II	slate 10.3%	10.4%	10.2%	10.8%	10.2%	10.2%	10.2%	10.5%	10.8%		
20Ozzie Albies	slate 10.0%	10.4%	10.0%	10.6%	10.6%	10.0%	10.0%	10.3%	10.2%		
21Freddie Freeman	slate 9.5%	9.2%	9.3%	9.2%	9.1%	9.3%	9.3%	9.3%	9.1%		
22Christian Yelich	slate 9.4%	9.6%	9.7%	9.3%	9.6%	9.7%	9.7%	9.8%	9.4%		
23Andy Pages	slate 9.2%	8.8%	8.9%	8.7%	9.0%	8.9%	8.9%	8.8%	8.7%		
24Xander Bogaerts	slate 9.0%	9.0%	8.8%	8.6%	8.8%	8.8%	8.8%	9.0%	8.7%		
25Luis Campusano	slate 7.7%	7.7%	7.9%	7.6%	7.8%	7.9%	7.9%	7.9%	7.6%		
26Enrique Hernández	slate 7.4%	7.3%	7.2%	6.6%	6.7%	7.2%	7.2%	7.0%	6.6%		
27Joey Ortiz	slate 7.3%	8.4%	7.5%	7.6%	7.5%	7.5%	7.5%	7.4%	7.4%		
28Kyle Tucker	slate 7.2%	7.3%	7.3%	7.1%	7.3%	7.3%	7.3%	7.2%	7.0%		

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
29Dustin HarrisSD	6.8%	8.2%	6.7%	5.8%	6.0%	6.7%	6.7%	6.6%	6.4%	
30Miguel RojasLAD	6.3%	7.8%	6.4%	6.7%	6.7%	6.4%	6.4%	6.7%	6.9%	
31Mauricio DubónTL	6.1%	7.4%	6.2%	6.5%	6.4%	6.2%	6.2%	6.0%	6.4%	
32Cooper PrattMIL	5.4%	7.0%	5.4%	4.9%	5.1%	5.4%	5.4%	4.9%	4.9%	
33Ha-Seong KimATL	5.2%	6.4%	4.5%	4.0%	3.9%	4.5%	4.5%	4.9%	4.8%	
34Sal FrelickMIL	4.9%	6.0%	4.8%	4.6%	4.6%	4.8%	4.8%	4.7%	4.7%	
35Jake Cronenworth	4.7%	5.9%	4.8%	4.8%	4.8%	4.8%	4.8%	4.8%	4.8%	
36David HamiltonMIL	4.3%	5.7%	4.5%	4.3%	4.0%	4.5%	4.5%	4.3%	4.3%	
37Luis LaraMIL	4.3%	6.0%	4.6%	3.7%	4.0%	4.6%	4.6%	3.6%	4.2%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.